

Teja

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AI/ML Engineer

Professional Summary

Production-focused AI/ML Engineer with 5+ years of experience delivering reliable production AI solutions across enterprise finance, healthcare, and technology environments. Proven expertise across Machine Learning Operations (MLOps), LLM orchestration (Python, PyTorch, LangChain, Docker, Kubernetes), and incident management (Agile/Scrum workflows). Highly proficient in modern deep learning frameworks, continuous deployment tooling, and generative AI agent architectures to deliver robust, scalable inference services and predictive insights.

Education

University of North Texas
Master of Science in Data Science

Denton, TX

Technical Skills

- **ML Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, CrewAI, LlamaIndex, NumPy, Pandas
- **Model Development & Evaluation:** Deep Learning (CNNs), Classification, Regression, Retrieval-Augmented Generation (RAG), Prompt Engineering, Drift Tracking, Telemetry, Groundedness Testing
- **MLOps & Deployment:** Docker, Kubernetes, Continuous Integration / Continuous Deployment (CI/CD), Git, FastAPI, RESTful APIs, Vertex AI, Canary/Shadow Deployments
- **Data Engineering & Pipelines:** Apache Spark (PySpark), SQL, NoSQL, Snowflake, AWS Glue, Extract Transform Load (ETL) Workflows
- **Cloud & Infrastructure:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, Amazon CloudWatch, S3, Athena, Azure Databricks
- **Methodologies & Operations:** Agile/Scrum, Incident Management, Service Level Objectives (SLOs) Definition, Root Cause Analysis (RCA)

Professional Experience

AI/ML Engineer, American Express

Jun 2024 – Present

- Engineered and deployed highly reliable enterprise AI services using Python and PyTorch, processing 2M+ daily transactions while meeting rigorous financial governance and security standards.
- Established operational readiness for production models by defining Service Level Objectives (SLOs), achieving 99.9% availability and maintaining p95 inference latency under 120ms.
- Orchestrated scalable Machine Learning Operations (MLOps) pipelines via Docker and Kubernetes, reducing model deployment time by 40% through automated CI/CD and canary deployment strategies.
- Integrated Retrieval-Augmented Generation (RAG) architectures with LangChain and internal knowledge bases, elevating query groundedness and response relevance by 35% for customer-facing workflows.
- Implemented comprehensive telemetry and data drift monitoring using Amazon CloudWatch, cutting incident triage time by 25% during production anomalies.
- Partnered directly with Data Scientists and frontend engineering teams in an Agile/Scrum environment to transition research prototypes into scalable, API-driven RESTful endpoints.

Machine Learning Engineer, GE Healthcare

Apr 2022 – Dec 2023

- Developed and validated deep learning architectures for predictive clinical diagnostics, accelerating model evaluation cycles by 30% across cross-functional engineering teams.
- Trained and fine-tuned complex neural networks using TensorFlow and PyTorch, achieving a 15% improvement in multi-class classification accuracy on 500GB+ of healthcare imagery.
- Built automated data engineering and Extract, Transform, Load (ETL) workflows to process diverse patient records, standardizing input schemas and improving data quality for downstream model consumption.
- Contributed to model explainability and auditability documentation, ensuring all deployed AI solutions complied with strict medical regulatory compliance requirements.
- Executed shadow deployments for updated inference models, validating performance consistency against legacy systems before routing 100% of production traffic.
- Designed robust automated testing frameworks (unit, integration, regression) for machine learning pipelines, preventing critical regressions across 15+ release cycles.

Data Scientist, Cyberspace Technologies

Jan 2021 – Mar 2022

- Transitioned experimental predictive models from prototype to production, delivering scalable analytics solutions that generated actionable insights for 10+ enterprise clients.
- Engineered large-scale data pipelines utilizing Apache Spark and SQL, processing 1TB+ of daily log telemetry to train robust regression and clustering algorithms.
- Executed advanced data wrangling and feature engineering workflows, elevating baseline model F1-scores by 12% during offline validation phases.
- Deployed predictive services into cloud-native architectures using Google Cloud Platform (GCP) Vertex AI, integrating REST APIs for seamless enterprise application consumption.
- Collaborated closely with platform teams to define input requirements, ensuring high availability and seamless data ingestion from distributed NoSQL databases.

Technical Projects

Multi-Agent Collaboration for Research Tasks

Python, CrewAI, LangChain, LlamaIndex, RAG, Prompt Engineering

- Developed an agentic AI system leveraging LangChain and CrewAI to orchestrate multi-step reasoning, planning, and task completion across diverse research domains.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using LlamaIndex and vector databases, optimizing context engineering and memory management to achieve 90%+ factual groundedness.

Enterprise Knowledge RAG Assistant for Information Systems

Python, LangChain, FastAPI, Docker, MLflow

- Designed and deployed a production-grade generative AI and predictive analytics platform, reducing complex information retrieval and query processing time by nearly 50%.
- Developed scalable FastAPI microservices while implementing ML-based query intent classification and confidence scoring to improve response reliability and decision support.
- Established end-to-end Continuous Integration / Continuous Deployment (CI/CD) and MLOps workflows with MLflow and Docker, incorporating monitoring dashboards for drift tracking and sustained accuracy.

AI-Driven Plant Disease Detection System

Python, PyTorch, TensorFlow, CNN, FastAPI, Deep Learning

- Built a scalable Computer Vision REST API using FastAPI and PyTorch, serving robust Deep Learning Convolutional Neural Networks (CNNs) for real-time disease classification.
- Applied robust data augmentation and preprocessing strategies to balance training distributions, ensuring highly reliable inference across diverse edge-case scenarios.

Spotify Analytics Platform for User Insights

PySpark, Databricks, AWS, Snowflake, SQL, ETL

- Architected an end-to-end cloud ETL pipeline on AWS using Amazon S3, AWS Glue, and PySpark running on Azure Databricks to process large-scale streaming data.
- Optimized complex SQL queries and data warehousing schemas in Snowflake, decreasing analytical dashboard refresh latency by 45% for business intelligence teams.

Certifications

- AWS Solution Architect Associate
- Azure AI Engineer Associate
- GCP Cloud Machine Learning Professional
- IBM Data Scientist Professional